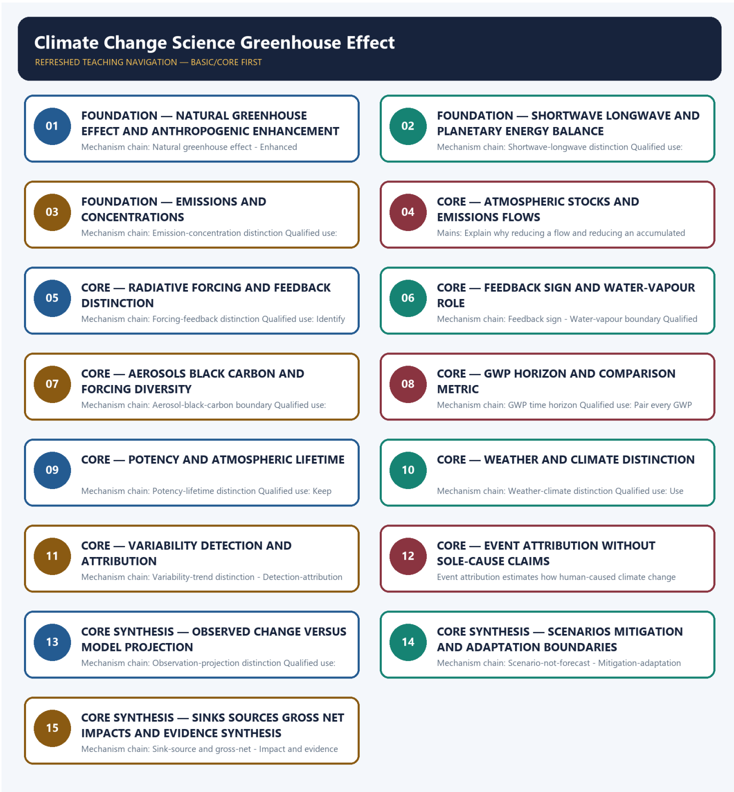
SOLVED PRACTICE WORKBOOK

Climate Change Science Greenhouse Effect - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Climate Change Science Greenhouse Effect - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Natural greenhouse effect?

A. The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. B. Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. C. An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. D. Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect.

Answer: A. Explanation: The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q2. Which option preserves the ecological boundary of Natural greenhouse effect?

A. Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. B. The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. C. An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. D. Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock.

Answer: B. Explanation: The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q3. Which statement uses Natural greenhouse effect without changing its scale, parameter or status?

A. An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. B. Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. C. The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. D. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change.

Answer: C. Explanation: The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q4. Which option avoids the standard UPSC close-option trap about Natural greenhouse effect?

A. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. B. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. C. Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. D. The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated.

Answer: D. Explanation: The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q5. Which statement correctly identifies Enhanced anthropogenic forcing?

A. Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. B. An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. C. Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. D. Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow.

Answer: A. Explanation: Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q6. Which option preserves the ecological boundary of Enhanced anthropogenic forcing?

A. An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. B. Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. C. Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. D. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change.

Answer: B. Explanation: Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q7. Which statement uses Enhanced anthropogenic forcing without changing its scale, parameter or status?

A. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. B. Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. C. Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. D. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change.

Answer: C. Explanation: Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q8. Which option avoids the standard UPSC close-option trap about Enhanced anthropogenic forcing?

A. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. B. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. C. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. D. Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect.

Answer: D. Explanation: Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q9. Which statement correctly identifies Shortwave-longwave distinction?

A. Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. B. Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. C. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. D. An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable.

Answer: A. Explanation: Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q10. Which option preserves the ecological boundary of Shortwave-longwave distinction?

A. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. B. Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. C. Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. D. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change.

Answer: B. Explanation: Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q11. Which statement uses Shortwave-longwave distinction without changing its scale, parameter or status?

A. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. B. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. C. Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. D. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable.

Answer: C. Explanation: Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q12. Which option avoids the standard UPSC close-option trap about Shortwave-longwave distinction?

A. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. B. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. C. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. D. Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow.

Answer: D. Explanation: Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q13. Which statement correctly identifies Emission-concentration distinction?

A. An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. B. Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. C. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. D. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change.

Answer: A. Explanation: An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q14. Which option preserves the ecological boundary of Emission-concentration distinction?

A. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. B. An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. C. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. D. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable.

Answer: B. Explanation: An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q15. Which statement uses Emission-concentration distinction without changing its scale, parameter or status?

A. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. B. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. C. An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. D. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable.

Answer: C. Explanation: An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q16. Which option avoids the standard UPSC close-option trap about Emission-concentration distinction?

A. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. B. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. C. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. D. An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable.

Answer: D. Explanation: An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q17. Which statement correctly identifies Stock-flow distinction?

A. Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. B. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. C. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. D. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change.

Answer: A. Explanation: Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q18. Which option preserves the ecological boundary of Stock-flow distinction?

A. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. B. Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. C. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. D. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing.

Answer: B. Explanation: Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q19. Which statement uses Stock-flow distinction without changing its scale, parameter or status?

A. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. B. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. C. Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. D. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign.

Answer: C. Explanation: Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q20. Which option avoids the standard UPSC close-option trap about Stock-flow distinction?

A. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. B. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. C. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. D. Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock.

Answer: D. Explanation: Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q21. Which statement correctly identifies Forcing-feedback distinction?

A. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. B. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. C. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. D. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing.

Answer: A. Explanation: Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q22. Which option preserves the ecological boundary of Forcing-feedback distinction?

A. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. B. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. C. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. D. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign.

Answer: B. Explanation: Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q23. Which statement uses Forcing-feedback distinction without changing its scale, parameter or status?

A. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. B. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. C. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. D. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign.

Answer: C. Explanation: Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q24. Which option avoids the standard UPSC close-option trap about Forcing-feedback distinction?

A. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. B. Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. C. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. D. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change.

Answer: D. Explanation: Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q25. Which statement correctly identifies Feedback sign?

A. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. B. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. C. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. D. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete.

Answer: A. Explanation: A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q26. Which option preserves the ecological boundary of Feedback sign?

A. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. B. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. C. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. D. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning.

Answer: B. Explanation: A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q27. Which statement uses Feedback sign without changing its scale, parameter or status?

A. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. B. Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. C. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. D. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning.

Answer: C. Explanation: A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q28. Which option avoids the standard UPSC close-option trap about Feedback sign?

A. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. B. Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. C. Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. D. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable.

Answer: D. Explanation: A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q29. Which statement correctly identifies Water-vapour boundary?

A. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. B. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. C. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. D. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete.

Answer: A. Explanation: Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q30. Which option preserves the ecological boundary of Water-vapour boundary?

A. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. B. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. C. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. D. Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend.

Answer: B. Explanation: Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q31. Which statement uses Water-vapour boundary without changing its scale, parameter or status?

A. Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. B. Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. C. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. D. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning.

Answer: C. Explanation: Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q32. Which option avoids the standard UPSC close-option trap about Water-vapour boundary?

A. Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. B. Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. C. Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. D. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing.

Answer: D. Explanation: Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. The other options belong to different media,

parameters, processes, scales, actors, institutions, instruments or status categories.

Q33. Which statement correctly identifies Aerosol-black-carbon boundary?

A. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. B. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. C. Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. D. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete.

Answer: A. Explanation: Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q34. Which option preserves the ecological boundary of Aerosol-black-carbon boundary?

A. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. B. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. C. Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. D. Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend.

Answer: B. Explanation: Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q35. Which statement uses Aerosol-black-carbon boundary without changing its scale, parameter or status?

A. Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. B. Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. C. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. D. Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency.

Answer: C. Explanation: Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q36. Which option avoids the standard UPSC close-option trap about Aerosol-black-carbon boundary?

A. Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. B. Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. C. Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. D. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign.

Answer: D. Explanation: Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q37. Which statement correctly identifies GWP time horizon?

A. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. B. Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. C. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. D. Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency.

Answer: A. Explanation: Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q38. Which option preserves the ecological boundary of GWP time horizon?

A. Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. B. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. C. Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. D. Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency.

Answer: B. Explanation: Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q39. Which statement uses GWP time horizon without changing its scale, parameter or status?

A. Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. B. Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. C. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. D. Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency.

Answer: C. Explanation: Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q40. Which option avoids the standard UPSC close-option trap about GWP time horizon?

A. Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. B. An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. C. Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. D. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete.

Answer: D. Explanation: Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q41. Which statement correctly identifies Potency-lifetime distinction?

A. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. B. Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. C. Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. D. Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence.

Answer: A. Explanation: Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q42. Which option preserves the ecological boundary of Potency-lifetime distinction?

A. Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. B. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. C. Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. D. Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change.

Answer: B. Explanation: Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q43. Which statement uses Potency-lifetime distinction without changing its scale, parameter or status?

A. Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. B. An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. C. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. D. Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence.

Answer: C. Explanation: Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments

or status categories.

Q44. Which option avoids the standard UPSC close-option trap about Potency-lifetime distinction?

A. A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. B. An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. C. Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. D. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning.

Answer: D. Explanation: Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q45. Which statement correctly identifies Weather-climate distinction?

A. Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. B. Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. C. Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. D. Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence.

Answer: A. Explanation: Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q46. Which option preserves the ecological boundary of Weather-climate distinction?

A. An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. B. Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. C. Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. D. Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change.

Answer: B. Explanation: Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q47. Which statement uses Weather-climate distinction without changing its scale, parameter or status?

A. A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. B. Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. C. Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. D. An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions.

Answer: C. Explanation: Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q48. Which option avoids the standard UPSC close-option trap about Weather-climate distinction?

A. An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. B. Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. C. A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. D. Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend.

Answer: D. Explanation: Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q49. Which statement correctly identifies Variability-trend distinction?

A. Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. B. Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. C. An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. D. Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change.

Answer: A. Explanation: Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q50. Which option preserves the ecological boundary of Variability-trend distinction?

A. Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. B. Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. C. A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. D. An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions.

Answer: B. Explanation: Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q51. Which statement uses Variability-trend distinction without changing its scale, parameter or status?

A. Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. B. A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. C. Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. D. An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions.

Answer: C. Explanation: Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q52. Which option avoids the standard UPSC close-option trap about Variability-trend distinction?

A. A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. B. Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. C. A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. D. Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency.

Answer: D. Explanation: Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q53. Which statement correctly identifies Detection-attribution distinction?

A. Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. B. A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. C. Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. D. An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions.

Answer: A. Explanation: Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q54. Which option preserves the ecological boundary of Detection-attribution distinction?

A. An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. B. Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. C. A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. D. Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action.

Answer: B. Explanation: Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. The other options belong to different media, parameters, processes, scales, actors,

institutions, instruments or status categories.

Q55. Which statement uses Detection-attribution distinction without changing its scale, parameter or status?

A. A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. B. A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. C. Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. D. Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action.

Answer: C. Explanation: Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q56. Which option avoids the standard UPSC close-option trap about Detection-attribution distinction?

A. Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. B. Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. C. A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. D. Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence.

Answer: D. Explanation: Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q57. Which statement correctly identifies Event-attribution boundary?

A. Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. B. A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. C. Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. D. An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions.

Answer: A. Explanation: Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q58. Which option preserves the ecological boundary of Event-attribution boundary?

A. A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. B. Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. C. A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. D. Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action.

Answer: B. Explanation: Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q59. Which statement uses Event-attribution boundary without changing its scale, parameter or status?

A. Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. B. A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. C. Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. D. Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action.

Answer: C. Explanation: Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q60. Which option avoids the standard UPSC close-option trap about Event-attribution boundary?

A. Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. B. The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. C. A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. D. Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change.

Answer: D. Explanation: Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q61. Which statement correctly identifies Observation-projection distinction?

A. An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. B. Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. C. A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. D. A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur.

Answer: A. Explanation: An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q62. Which option preserves the ecological boundary of Observation-projection distinction?

A. Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. B. An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. C. A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. D. Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale.

Answer: B. Explanation: An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q63. Which statement uses Observation-projection distinction without changing its scale, parameter or status?

A. Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. B. The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. C. An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. D. A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate.

Answer: C. Explanation: An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q64. Which option avoids the standard UPSC close-option trap about Observation-projection distinction?

A. The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. B. Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. C. Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. D. An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions.

Answer: D. Explanation: An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q65. Which statement correctly identifies Scenario-not-forecast?

A. A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. B. A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. C. Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. D. Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale.

Answer: A. Explanation: A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q66. Which option preserves the ecological boundary of Scenario-not-forecast?

A. A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. B. A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. C. Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. D. The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated.

Answer: B. Explanation: A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q67. Which statement uses Scenario-not-forecast without changing its scale, parameter or status?

A. Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. B. The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. C. A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. D. Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale.

Answer: C. Explanation: A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q68. Which option avoids the standard UPSC close-option trap about Scenario-not-forecast?

A. The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. B. Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. C. Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. D. A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur.

Answer: D. Explanation: A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q69. Which statement correctly identifies Mitigation-adaptation distinction?

A. Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. B. The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. C. A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. D. Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale.

Answer: A. Explanation: Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q70. Which option preserves the ecological boundary of Mitigation-adaptation distinction?

A. Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. B. Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. C. Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. D. The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated.

Answer: B. Explanation: Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q71. Which statement uses Mitigation-adaptation distinction without changing its scale, parameter or status?

A. Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. B. The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. C. Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. D. Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow.

Answer: C. Explanation: Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q72. Which option avoids the standard UPSC close-option trap about Mitigation-adaptation distinction?

A. Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. B. An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. C. Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. D. Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action.

Answer: D. Explanation: Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q73. Which statement correctly identifies Sink-source and gross-net?

A. A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. B. Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. C. The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. D. Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect.

Answer: A. Explanation: A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q74. Which option preserves the ecological boundary of Sink-source and gross-net?

A. The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. B. A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. C. Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. D. Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect.

Answer: B. Explanation: A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q75. Which statement uses Sink-source and gross-net without changing its scale, parameter or status?

A. Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. B. An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. C. A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. D. Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect.

Answer: C. Explanation: A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q76. Which option avoids the standard UPSC close-option trap about Sink-source and gross-net?

A. Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. B. An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. C. Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. D. A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate.

Answer: D. Explanation: A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q77. Which statement correctly identifies Impact and evidence boundary?

A. Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. B. The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. C. Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. D. Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow.

Answer: A. Explanation: Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q78. Which option preserves the ecological boundary of Impact and evidence boundary?

A. Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. B. Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. C. Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. D. An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable.

Answer: B. Explanation: Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q79. Which statement uses Impact and evidence boundary without changing its scale, parameter or status?

A. Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. B. An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. C. Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. D. Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation;

greenhouse gases interact with the outgoing longwave part of this energy flow.

Answer: C. Explanation: Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q80. Which option avoids the standard UPSC close-option trap about Impact and evidence boundary?

A. An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. B. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. C. Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. D. Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale.

Answer: D. Explanation: Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

PYQS AND ANSWER PRACTICE

AUDITED CLIMATE-SCIENCE, ATTRIBUTION, IMPACT AND RESPONSE PYQ OWNERSHIP

Audited ledgers route objective demands on carbon fertilisation, geoengineering, methane hydrates, agricultural gases, rice practices, cement emissions and India's carbon-dioxide profile, plus verified Mains demands on greenhouse-gas effects, tropical food security and island sea-level risk. Objective keys are not inferred.

OWNER PYQ LEDGER EXTRACTS

9. PYQ application

- ANALYSIS: Recurring Prelims pattern: identify the correct greenhouse gases, their relative

potency (GWP) and atmospheric lifetime characteristics.

- ANALYSIS: Mains linkage: the natural-versus-enhanced greenhouse-effect distinction is used to

precisely frame climate-change causation in policy-analysis answers.

Recent PYQ Integration (2024-2025)

> **Status:** 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2024-2025.md`, `_PYQ-ROUTING-PRELIMS-2024-2025.md`. **Answer-key rule:** The official 2024-2025 Prelims Set-A keys are present in the repository and CSAT Set-A keys are supplied; even so, no option or answer is recorded or inferred in this integration.

- **Years represented:** 2025
- **Paper(s):** GS-I, Prelims GS-I
- **Routed question demands:** 3

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2025	GS-I	4	Climate change and sea-level rise affecting island nations	Discuss · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2025	Prelims GS-I	31	Cement-industry carbon emissions; limestone and clinker	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2025	Prelims GS-I	38	India's CO2 emissions per capita and largest sources	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Climate change and sea-level rise affecting island nations
- Cement-industry carbon emissions; limestone and clinker
- India's CO2 emissions per capita and largest sources

> This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

Historical PYQ Integration (2018-2023)

> **Status:** Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2018-2023.md`, `_PYQ-ROUTING-MAINS-GS3-GS4-2018-2023.md`, `_PYQ-ROUTING-PRELIMS-2018-2023.md`. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2019, 2022, 2023
- **Paper(s):** GS-I, GS-III, Prelims GS-I
- **Routed question demands:** 7

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	Prelims GS-I	65	Carbon fertilization effect on plant growth and atmosphere	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2019	Prelims GS-I	25	Geoengineering techniques to counter global warming effects	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2019	Prelims GS-I	34	Methane hydrate deposits characteristics and climate linkage	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2022	GS-III	17	Global warming greenhouse gas effects and Kyoto Protocol measures	Discuss · 15 marks · 250 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2022	Prelims GS-I	21	Crops as anthropogenic sources of methane and nitrous oxide	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2022	Prelims GS-I	22	System of Rice Intensification methane reduction and inputs	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2023	GS-I	4	Climate change and food security in tropical countries	Discuss the consequences · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

What this owner must now support

- Carbon fertilization effect on plant growth and atmosphere
- Geoengineering techniques to counter global warming effects
- Methane hydrate deposits characteristics and climate linkage
- Global warming greenhouse gas effects and Kyoto Protocol measures
- Crops as anthropogenic sources of methane and nitrous oxide
- System of Rice Intensification methane reduction and inputs
- Climate change and food security in tropical countries

> The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

10. PYQ-based analytical application

- ANALYSIS: Prelims questions on GWP/atmospheric lifetime comparisons across gases should be answered using the potency-versus-persistence trade-off framework (Section 2 table).
- ANALYSIS: Mains answers on "climate science and policy" should explicitly invoke the carbon-budget-depletion framing and the aerosol-masking nuance to demonstrate quantitative and mechanistic understanding beyond descriptive greenhouse-effect explanation.

Historical PYQ Integration (2018-2023)

> **Status:** Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2018-2023.md`, `_PYQ-ROUTING-MAINS-GS3-GS4-2018-2023.md`.

- **Years represented:** 2022, 2023
- **Paper(s):** GS-I, GS-III
- **Routed question demands:** 2

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2022	GS-III	17	Global warming greenhouse gas effects and Kyoto Protocol measures	Discuss · 15 marks · 250 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2023	GS-I	4	Climate change and food security in tropical countries	Discuss the consequences · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

What this owner must now support

- Global warming greenhouse gas effects and Kyoto Protocol measures

- Climate change and food security in tropical countries

> The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

PYQ DEMAND CARD 1 - 2022 GS-III

Demand: Discuss global-warming greenhouse-gas effects before linking measures under Kyoto.

Status: Verified routed Mains demand; treaty measures are cross-owned by Topic 19.

Model solution: Natural greenhouse effect: The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. **Enhanced anthropogenic forcing:** Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. **Emission-concentration distinction:** An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. **Forcing-feedback distinction:** Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. **GWP time horizon:** Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. **Mitigation-adaptation distinction:** Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

PYQ DEMAND CARD 2 - 2023 GS-I

Demand: Discuss consequences of climate change for food security in tropical countries.

Status: Verified routed Mains demand; no official model answer is claimed.

Model solution: Weather-climate distinction: Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. **Variability-trend distinction:** Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. **Detection-attribution distinction:** Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. **Observation-projection distinction:** An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. **Mitigation-adaptation distinction:** Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. **Impact and evidence boundary:** Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

PYQ DEMAND CARD 3 - 2025 GS-I

Demand: Discuss climate change and sea-level rise affecting island nations.

Status: Verified routed Mains demand; no local projection figure is inferred.

Model solution: Detection-attribution distinction: Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. **Observation-projection distinction:** An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. **Scenario-not-forecast:** A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. **Mitigation-adaptation distinction:** Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. **Impact and evidence boundary:** Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish the natural greenhouse effect from enhanced anthropogenic forcing. Answer in about 150 words.

Model thesis: Claim: Natural greenhouse effect. **Named evidence/example:** The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Enhanced anthropogenic forcing. **Named evidence/example:** Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Shortwave-longwave distinction. **Named evidence/example:** Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated.
- Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect.
- Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow.

Qualified conclusion: Claim: Natural greenhouse effect. **Named evidence/example:** The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Enhanced anthropogenic forcing. **Named evidence/example:** Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Shortwave-longwave distinction. **Named evidence/example:** Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 2 - 10 MARKS

Question: Differentiate emissions, concentrations, stocks and flows in climate accounting. Answer in about 150 words.

Model thesis: Claim: Emission-concentration distinction. **Named evidence/example:** An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Stock-flow distinction. **Named evidence/example:** Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Sink-source and gross-net. **Named evidence/example:** A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable.
- Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock.
- A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate.

Qualified conclusion: Claim: Emission-concentration distinction. **Named evidence/example:** An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Stock-flow distinction. **Named evidence/example:** Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Sink-source and gross-net. **Named evidence/example:** A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain radiative forcing, feedbacks and gas-specific climate metrics. Answer in about 250 words.

Model thesis: Claim: Forcing-feedback distinction. **Named evidence/example:** Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Feedback sign. **Named evidence/example:** A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Water-vapour boundary. **Named evidence/example:** Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Aerosol-black-carbon boundary. **Named evidence/example:** Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** GWP time horizon. **Named evidence/example:** Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Potency-lifetime distinction. **Named evidence/example:** Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change.
- A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable.
- Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing.
- Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign.
- Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete.
- Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning.

Qualified conclusion: Claim: Forcing-feedback distinction. **Named evidence/example:** Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Feedback sign. **Named evidence/example:** A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Water-vapour boundary. **Named evidence/example:** Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Aerosol-black-carbon boundary. **Named evidence/example:** Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** GWP time horizon. **Named evidence/example:** Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Potency-lifetime distinction. **Named evidence/example:** Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 4 - 15 MARKS

Question: Distinguish climate trend, detection, attribution and event attribution. Answer in about 250 words.

Model thesis: Claim: Weather-climate distinction. **Named evidence/example:** Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Variability-trend distinction. **Named evidence/example:** Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Detection-attribution distinction. **Named evidence/example:** Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Event-attribution boundary. **Named evidence/example:** Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend.
- Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency.
- Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence.
- Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change.

Qualified conclusion: Claim: Weather-climate distinction. **Named evidence/example:** Weather describes short-period atmospheric conditions, whereas climate concerns statistical patterns over longer periods; one unusual season or event does not alone establish a climate trend. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Variability-trend distinction. **Named evidence/example:** Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Detection-attribution distinction. **Named evidence/example:** Detection asks whether an observed change is distinguishable from expected variability, while attribution evaluates the relative contributions of human and natural drivers using multiple lines of evidence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Event-attribution boundary. **Named evidence/example:** Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate how climate science should translate into mitigation and adaptation policy. Answer in about 300 words.

Model thesis: Claim: Enhanced anthropogenic forcing. **Named evidence/example:** Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Emission-concentration distinction. **Named evidence/example:** An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Forcing-feedback distinction. **Named evidence/example:** Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** GWP time horizon. **Named evidence/example:** Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Potency-lifetime distinction. **Named evidence/example:** Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Observation-projection distinction. **Named evidence/example:** An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Scenario-not-forecast. **Named evidence/example:** A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Mitigation-adaptation distinction. **Named evidence/example:** Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect.
- An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable.
- Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change.
- Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete.
- Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning.
- An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions.
- A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur.
- Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action.

Qualified conclusion: Claim: Enhanced anthropogenic forcing. **Named evidence/example:** Current climate concern arises from human activities increasing greenhouse-gas concentrations and other forcing agents, thereby altering the climate system beyond the natural greenhouse effect. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Emission-concentration distinction. **Named evidence/example:** An emission is a flow released during a period, while atmospheric concentration is the resulting abundance after sources, sinks, chemistry and lifetime act; the two quantities are related but not interchangeable. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Forcing-feedback distinction. **Named evidence/example:** Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** GWP time horizon. **Named evidence/example:** Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Potency-lifetime distinction. **Named evidence/example:** Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Observation-projection distinction. **Named evidence/example:** An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Scenario-not-forecast. **Named evidence/example:** A climate scenario is a coherent conditional pathway used to explore possible futures; it is not an unconditional prediction that a particular future will occur. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Mitigation-adaptation distinction. **Named evidence/example:** Mitigation addresses sources or enhances sinks to limit climate change, whereas adaptation adjusts human or natural systems to actual or expected impacts; neither term is a synonym for all climate action. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 6 - 20 MARKS

Question: Build an evidence-safe climate-change answer from mechanism to impacts. Answer in about 300 words.

Model thesis: Claim: Natural greenhouse effect. **Named evidence/example:** The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Shortwave-longwave distinction. **Named evidence/example:** Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Stock-flow distinction. **Named evidence/example:** Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Feedback sign. **Named evidence/example:** A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Aerosol-black-carbon boundary. **Named evidence/example:** Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Variability-trend distinction. **Named evidence/example:** Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Event-attribution boundary. **Named evidence/example:** Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Observation-projection distinction. **Named evidence/example:** An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Sink-source and gross-net. **Named evidence/example:** A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Impact and evidence boundary. **Named evidence/example:** Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated.
- Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow.
- Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock.
- A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable.
- Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign.
- Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency.
- Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change.
- An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions.
- A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate.
- Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale.

Qualified conclusion: Claim: Natural greenhouse effect. **Named evidence/example:** The natural greenhouse effect is a life-enabling energy-balance process in which greenhouse gases absorb and re-emit part of Earth's outgoing longwave radiation; it is not the anthropogenic problem to be eliminated. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Shortwave-longwave distinction. **Named evidence/example:** Incoming solar energy is predominantly shortwave, whereas the warmed surface emits longwave infrared radiation; greenhouse gases interact with the outgoing longwave part of this energy flow. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Stock-flow distinction. **Named evidence/example:** Long-lived greenhouse gases create a stock problem because accumulated past and present emissions influence concentration; lowering an annual flow does not by itself remove the existing atmospheric stock. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Feedback sign. **Named evidence/example:** A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Aerosol-black-carbon boundary. **Named evidence/example:** Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Variability-trend distinction. **Named evidence/example:** Natural variability can raise or lower conditions around a long-term trend, so short-term fluctuation neither proves nor disproves the underlying climate tendency. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Event-attribution boundary. **Named evidence/example:** Event attribution estimates how human-caused climate change altered the probability or intensity of a defined event class; it does not justify saying that every individual disaster was caused solely by climate change. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Observation-projection distinction. **Named evidence/example:** An observation describes measured past or present change, whereas a model projection is a conditional statement about a future pathway under specified assumptions. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Sink-source and gross-net. **Named evidence/example:** A sink absorbs more of a substance than it releases over the stated boundary and period, while a source releases more; gross removals, gross emissions and net balance must be kept separate. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Impact and evidence boundary. **Named evidence/example:** Ocean acidification follows carbon-dioxide uptake and chemistry, while warming and sea-level impacts involve additional mechanisms; global findings, India-specific evidence, observed attribution and future projections must be cited at their proper scale. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.